THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS CHAPTER TEST SET A

Chapter 3: **Pair of Linear Equations in Two Variables** | RBSE Board Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Use graph paper for Q. 18 (if attempted).

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. The pair of equations $x + 2y = 3$ and $2x + 4y = 6$ has: 1

(a) a unique solution **(b)** no solution
(c) infinitely many solutions **(d)** exactly two solutions

2. The lines represented by $2x + 3y = 7$ and $4x + 6y = 9$ are: 1

(a) intersecting **(b)** parallel **(c)** coincident **(d)** perpendicular

3. The pair $kx + 2y = 5$ and $3x + y = 1$ has a unique solution if: 1

(a) $k = 6$ **(b)** $k \neq 6$ **(c)** $k = 3$ **(d)** $k \neq 3$

4. Which point lies on the line $2x + 3y = 12$? 1

(a) $(3, 2)$ **(b)** $(2, 3)$ **(c)** $(0, 3)$ **(d)** $(6, 1)$

5. A student states: "If $\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$, the pair of equations is inconsistent." The statement is: 1

(a) Correct
(b) Incorrect — the pair has infinitely many solutions and is consistent
(c) Correct only if $c_1 = 0$
(d) Cannot be determined

6. The solution of $x + y = 5$ and $x - y = 1$ is: 1

(a) $(2, 3)$ **(b)** $(3, 2)$ **(c)** $(4, 1)$ **(d)** $(1, 4)$

7. For $3x - y = 3$ and $9x - 3y = 9$, Student A says "no solution" and Student B says "infinitely many solutions". Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

8. If $x = 2,\ y = 1$ is a solution of $2x + ky = 7$, then k is: 1

(a) 1 **(b)** 2 **(c)** 3 **(d)** 4

9. The graph of the equation $x = 4$ is a line: 1

(a) parallel to the x-axis **(b)** parallel to the y-axis
(c) passing through the origin **(d)** the y-axis itself

10. The sum of two numbers is 20 and their difference is 4. The numbers are: 1

(a) $12, 8$ **(b)** $14, 6$ **(c)** $16, 4$ **(d)** $10, 10$

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Solve by the substitution method: $x + y = 14,\ x - y = 4$. 2

12. Find the value of k for which the pair $x + 2y = 5$ and $3x + ky + 15 = 0$ has no solution. 2

13. Check whether the pair $\frac{3}{2}x + \frac{5}{3}y = 7$ and $9x - 10y = 14$ is consistent. 2

14. Half the perimeter of a rectangular garden, whose length is 4 m more than its width, is 36 m. Find the dimensions of the garden. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Solve by the elimination method: $3x + 4y = 10$ and $2x - 2y = 2$. 3

16. 5 pencils and 7 pens together cost ₹50, whereas 7 pencils and 5 pens together cost ₹46. Find the cost of one pencil and one pen. 3

17. Five years hence, the age of Jacob will be three times that of his son. Five years ago, Jacob's age was seven times that of his son. What are their present ages? 3

OR

A fraction becomes $\frac{1}{3}$ when 1 is subtracted from the numerator, and it becomes $\frac{1}{4}$ when 8 is added to its denominator. Find the fraction. 3

18. Draw the graphs of $x - y + 1 = 0$ and $3x + 2y - 12 = 0$. Determine the coordinates of the vertices of the triangle formed by these lines and the x-axis. 3

OR

Solve graphically: $x + 3y = 6$ and $2x - 3y = 12$. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. **Case study:** A taxi charges a fixed amount plus a charge per km. For a 10 km journey the fare is ₹105 and for a 15 km journey it is ₹155. 5

(i) Form a pair of linear equations.

(ii) Find the fixed charge and the charge per km.

(iii) What will a person pay for 25 km?

(iv) A student says a 0 km booking costs ₹0. Justify or refute.

OR

A part of the monthly hostel charges is fixed and the rest depends on the number of days one takes food in the mess. Student A takes food for 20 days and pays ₹1000; Student B takes food for 26 days and pays ₹1180. Find the fixed charge and the cost of food per day. How much will Student C pay for 30 days? 5

20. The sum of the digits of a two-digit number is 9. Also, nine times this number is twice the number obtained by reversing the order of the digits. Find the number and verify your answer. 5

OR

Meena went to a bank to withdraw ₹2000. She asked the cashier to give her ₹50 and ₹100 notes only and got 25 notes in all. 5

(i) Find how many notes of each kind she received.

(ii) If she had asked for ₹2000 in 30 such notes, how many of each would she get?

END OF QUESTION PAPER — Total: 40 Marks